

PRECIS-2 Inter-Rater Reliability Analysis

Aarian Bhakoo

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This report reproduces the prespecified inter-rater reliability analyses for the finalized set of 237 paired PRECIS-2 ratings. Codes 1–5 represent the ordinal pragmatism scale; code 6 represents Missing and code 7 represents Not Applicable. The archived statistician-reviewed implementation is retained unchanged, and its Krippendorff's alpha machinery is preserved below.

1 Setup

2 Defining Agreement

3 Loading Files

4 Validation

```
## Setup completed successfully.
```

```
## Number of raters: 2
```

```
## Number of rated observations: 237
```

```
## Observed main-file codes: 1, 2, 3, 4, 5, 6, 7
```

5 Crude Agreement

Table 1: Summary of crude agreement under each agreement definition.

Definition	Agreements	Total	Crude_Agreement
Total	115	237	0.4852
Range	165	237	0.6962
Censored	110	211	0.5213
Buckets	152	237	0.6414

6 Krippendorff's Alpha

7 Disagreement Function

8 Four Definitions of Agreement

9 Dataset Transformation & Validation

10 Wrapper

11 Calls to Calculate Krippendorff's Alpha

Table 2: Summary of Krippendorff’s alpha under each agreement definition.

Definition	Observations	Alpha	Lower_95_CI	Upper_95_CI
Total	237	0.3005	0.1717	0.4193
Range	237	0.3350	0.1805	0.4734
Censored	211	0.3391	0.2039	0.4617
Bucketed	237	0.2471	0.1133	0.3721

12 Secondary Analyses

12.1 Crude Agreement by Domain

Table 3: Crude agreement by PRECIS-2 domain and agreement definition.

domain	Definition	Agreements	Total	Crude_Agreement
Eligibility	Total	13	24	0.5417
Eligibility	Range	17	24	0.7083
Eligibility	Censored	12	22	0.5455
Eligibility	Bucketed	15	24	0.6250
Recruitment	Total	13	24	0.5417
Recruitment	Range	18	24	0.7500
Recruitment	Censored	10	17	0.5882
Recruitment	Bucketed	17	24	0.7083
Setting	Total	13	24	0.5417
Setting	Range	19	24	0.7917
Setting	Censored	13	24	0.5417
Setting	Bucketed	17	24	0.7083
Organization	Total	10	24	0.4167
Organization	Range	17	24	0.7083
Organization	Censored	10	24	0.4167
Organization	Bucketed	16	24	0.6667
Flexibility in Delivery	Total	16	24	0.6667
Flexibility in Delivery	Range	20	24	0.8333
Flexibility in Delivery	Censored	16	24	0.6667
Flexibility in Delivery	Bucketed	20	24	0.8333
Flexibility in Adherence	Total	10	24	0.4167
Flexibility in Adherence	Range	11	24	0.4583
Flexibility in Adherence	Censored	10	15	0.6667
Flexibility in Adherence	Bucketed	11	24	0.4583
Follow-up	Total	12	23	0.5217
Follow-up	Range	17	23	0.7391
Follow-up	Censored	12	22	0.5455
Follow-up	Bucketed	14	23	0.6087
Primary Outcome	Total	10	23	0.4348
Primary Outcome	Range	15	23	0.6522
Primary Outcome	Censored	10	23	0.4348
Primary Outcome	Bucketed	13	23	0.5652
Primary Analysis	Total	7	23	0.3043

domain	Definition	Agreements	Total	Crude_Agreement
Primary Analysis	Range	10	23	0.4348
Primary Analysis	Censored	6	16	0.3750
Primary Analysis	Bucketed	8	23	0.3478
Comparator	Total	11	24	0.4583
Comparator	Range	21	24	0.8750
Comparator	Censored	11	24	0.4583
Comparator	Bucketed	21	24	0.8750

12.2 Mean Signed Deviation from Consensus

Table 4: Reviewer deviation from consensus after excluding codes 6 and 7.

Reviewer	Observations	Signed_Deviation	Absolute_Deviation	Mean Signed_Deviation	Mean Absolute_Deviation
Aarian	211	-3	97	-0.01422	0.4597
Merrick	211	6	92	0.02844	0.4360

13 Reproducibility Information

```
## R version 4.3.1 (2023-06-16 ucrt)
## Platform: x86_64-w64-mingw32/x64 (64-bit)
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##
## locale:
## [1] LC_COLLATE=English_Canada.utf8  LC_CTYPE=English_Canada.utf8    LC_MONETARY=English_Canada.utf8
##
## time zone: America/Toronto
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] krippendorffsalph_2.0 knitr_1.43      tibble_3.2.1    purrr_1.0.2     tidy
##
## loaded via a namespace (and not attached):
## [1] vctrs_0.6.5      cli_3.6.1      rlang_1.1.1     xfun_0.39       generics_0.1.3  glue_
## [17] digest_0.6.32    R6_2.5.1       tidyselect_1.2.1 utf8_1.2.4      pillar_1.9.0    magri
```